

# Bayram Bayramov

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## Profile

AI/ML Engineer with strong theoretical foundations and hands-on experience building production systems, computer vision models, and NLP applications. Built a production-ready AI food analyzer with 190+ tests and Docker deployment, implemented 7 ML algorithms from scratch, fine-tuned YOLO for object detection, and trained SAR segmentation models in PyTorch. Experienced in teaching Python and ML frameworks. Currently focused on improving MLOps and deployment skills with a practical approach to problem-solving.

## Skills

- Languages:** Python
- ML/DL:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas
- Computer Vision:** OpenCV, YOLO, U-Net, Attention U-Net, SAR segmentation
- NLP:** LSTM, Transformers, Hugging Face, NLTK
- Backend and APIs:** FastAPI, Pydantic, asyncpg
- Databases:** PostgreSQL, SQLite
- MLOps:** Docker, Kubernetes (Kind), Helm, GitHub Actions
- Testing:** Pytest, pytest-cov, pytest-asyncio

## Experience

- Azercosmos**, Machine Learning Intern

  - Built and trained SAR image segmentation models including U-Net, Attention U-Net, and LRAU-Net in PyTorch, working with 50+ GB of satellite imagery data
  - Diagnosed model failures through systematic error analysis including failure-case visualization, per-region statistics, and NDVI/NDBI indices
  - Conducted ablation experiments across architectures, developing a practical understanding of when ML approaches are appropriate versus simpler computer vision methods
- Baku, Azerbaijan  
Mar 2026 – May 2026  
3 months
- Div Academy**, Teacher and Tutor

  - Taught Python, SQL, PyTorch, TensorFlow, FastAPI, and Hugging Face Transformers to 30+ students across 3 cohorts
  - Guided students through end-to-end ML projects including regression, classification, unsupervised learning, NLP pipelines, and computer vision applications
  - Provided code review and technical mentorship, helping students debug model training and improve implementation quality
- Baku, Azerbaijan  
Oct 2025 – Mar 2026  
6 months
- Kapital Bank**, Data Science and AI Intern

  - Prepared and validated audio datasets for ASR model testing, working within a 4-person labeling team
  - Assisted in standardizing data preparation workflows to improve consistency for model evaluation
- Baku, Azerbaijan  
Aug 2025 – Aug 2025  
1 month
- DataMinds Azercell**, Data Analytics and AI Bootcamp

  - Built a RAG application using pgvector and LLM frameworks during an intensive bootcamp
  - Gained exposure to MLOps concepts including pipeline orchestration, containerization, and stream processing
- Baku, Azerbaijan  
July 2025 – Aug 2025  
2 months

## Projects

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### AI Food Analyzer

- Built a production-ready web application using FastAPI that identifies ingredients from meal photos and calculates nutritional information via Vision Language Models with automatic failover across OpenAI Anthropic and Google Gemini
- Implemented 3-layer caching achieving 500-600x speed improvement and parallel nutrition lookups with asyncio for up to 13x faster processing
- Containerized with Docker and Docker Compose using multi-stage builds with GitHub Actions CI achieving 86 percent test coverage across 190 tests
- Integrated PostgreSQL with asyncpg for persistent storage and built both REST API and CLI interfaces

### Ferocious Minsky ML Library from Scratch

- Implemented 7 machine learning algorithms from scratch using NumPy including Decision Trees AdaBoost Random Forest with OOB scoring and parallel training PCA K-Means DBSCAN and Gradient Boosting with Newton-Raphson updates
- Achieved 124 tests with 75 percent coverage verified against scikit-learn equivalents
- Built experiment framework for bias-variance decomposition and head-to-head algorithm comparison

### RAG Pipeline with Kubernetes HAProxy and Haystack

- Deployed RAG system on Kind Kubernetes with Qdrant vector DB Ollama TinyLLaMA and HAProxy
- Built Haystack indexing and retrieval pipeline for document collection
- Configured persistent volumes resource limits and readiness and liveness probes

### CI/CD Pipeline for FastAPI Service

- Designed GitHub Actions CI/CD pipeline with Docker build and push and Helm deployment to Kind cluster
- Implemented multi-stage Docker builds with non-root user and HEALTHCHECK
- Deployed FastAPI service to Kubernetes with resource limits and health probes

### Seismic First-Break Picking

- Built a deep learning pipeline for automated first-break picking on 2D seismic shot gathers, framing the task as 3-class segmentation and extracting per-trace picks from the first-break strip
- Trained and benchmarked 9 U-Net variants from 2K to 31M parameters on 690 shots, using a chunk-aware sampler, learning rate scheduling, and early stopping on validation IoU
- Achieved a pick median error of 1 sample and an accuracy of 74.96 percent within 3 samples, outperforming the classical STA LTA baseline which scored 3 samples and 51.9 percent
- Built evaluation and explainability tooling including Grad-CAM heatmaps, wavefront coherence analysis, and worst-N error galleries to diagnose failure modes
- Packaged inference with TorchScript and ONNX export verified for numeric equivalence, plus an MLflow tracked training and evaluation pipeline

## Education

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### Bachelor - Baku Higher Oil School, Chemical Engineering

- Built a strong foundation in mathematics, physics, and computational methods through a rigorous engineering curriculum
- Developed analytical problem-solving skills applicable to data science and modeling

Baku, Azerbaijan  
Sept 2020 – June 2025

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|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Certificate Program</b> | <b>AI Academy</b> , Artificial Intelligence <ul style="list-style-type: none"> <li>Studying machine learning and deep learning with a theory-heavy curriculum, implementing concepts through hands-on coding projects</li> <li>Built production-ready AI applications and ML libraries from scratch, with full implementations on GitHub</li> </ul> | Baku, Azerbaijan<br>Aug 2025 – Aug 2026 |
| <b>Master</b>              | <b>Azerbaijan State University of Economics</b> , Business and Data Analytics <ul style="list-style-type: none"> <li>Pursuing a Master’s degree combining data science with business applications</li> <li>Coursework includes advanced analytics, data visualization, and quantitative methods</li> </ul>                                          | Baku, Azerbaijan<br>Sept 2025 – present |